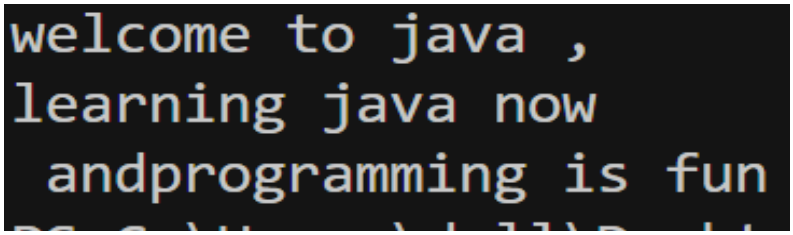
PROBLEM:-1 Write a program that display Welcome to java,Learning Java Now and Programming is Fun

CODE:-
```
import java.util.*;
class welcome{
    public static void main(String[] args) {
        System.out.println("welcome to java , \nlearning java now\n and programming is fun");
    }
}
```

OUTPUT:-



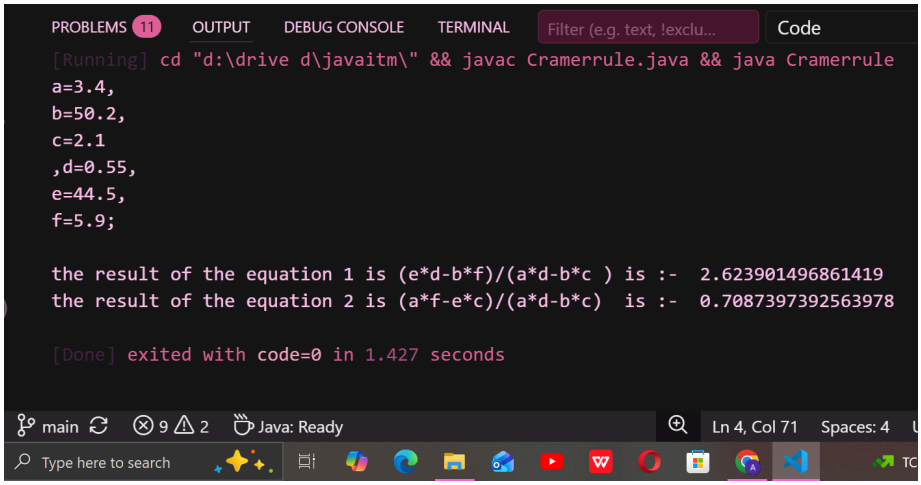
PROBLEM:-2 Write a program that solves the following equation and displays the value x and y:
1)3.4x+50.2y=44.5 2)2.1x+55y=5.9(assume cramer's rule to solve equation ax+by=e and cx+dy=f)
Where x=ed-bf/ad-bc y=af-ec/ad-bc

CODE:-
```
public class Cramerrule{
    public static void main(String[] args) {
        double a=3.4,b=50.2,c=2.1,d=0.55,e=44.5,f=5.9;
        System.out.println("a=3.4,\nb=50.2,\nc=2.1\n,d=0.55,\ne=44.5,\nf=5.9;\n");
        double x= (e*d-b*f)/(a*d-b*c ) , y=(a*f-e*c)/(a*d-b*c);
        System.out.println("the result of the equation 1 is (e*d-b*f)/(a*d-b*c ) is :- "+x);
        System.out.println("the result of the equation 2 is (a*f-e*c)/(a*d-b*c)  is :- "+y);

    }

}
```

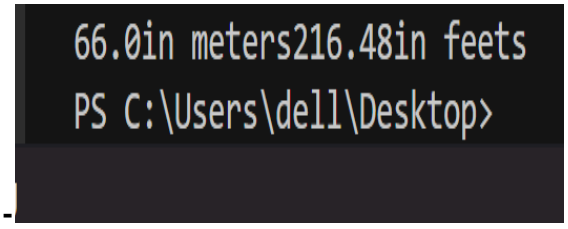
OUTPUT:-



PROBLEM:-3 Write a program that reads a number in meters,converts in feet, and displays the result

CODE:-

```
public class prac2itm {
    public static void main(String[] args) {
        double num=66;
        double result;
        result=num*3.28;
        System.out.println(num+"in meters"+result+"in feets");
    }
}
```



OUTPUT:-

PROBLEM:-4

CODE:-

OUTPUT:-

PROBLEM:-5

CODE:-

OUTPUT:-

PROBLEM:-6

CODE:-

OUTPUT:-

Practical:-4

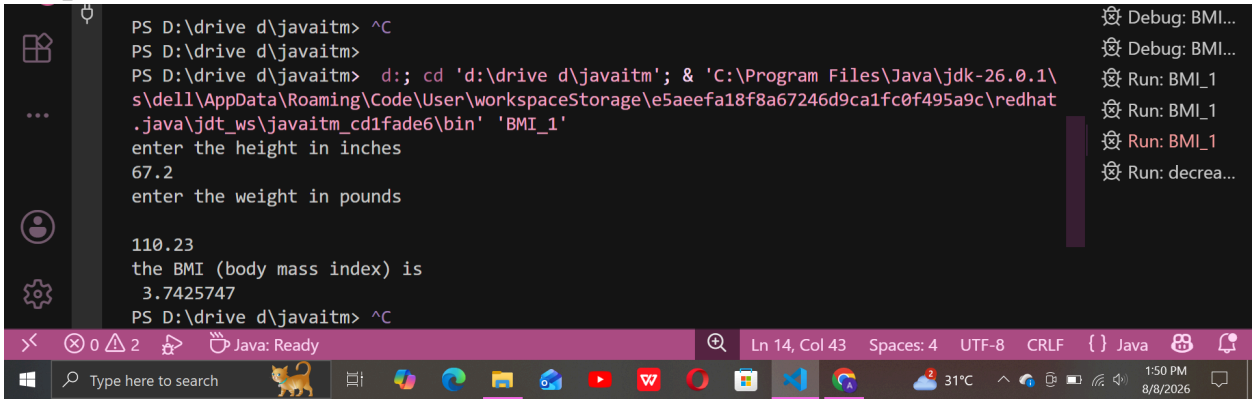
Problem Statement:-

Code:-

```
import java.util.Scanner;

public class BMI_1 {
    public static void main(String[] args) {
        float height;
        float bmi;
        float weight;
        Scanner scanner  = new Scanner(System.in);
        System.out.println("enter the height in inches \n ");
        height = scanner.nextFloat();
        System.out.println("enter the weight in pounds \n ");
        weight = scanner.nextFloat();
        float weightinkg = weight*4535927;
        float heightinmeter = height*0254;
        bmi = weightinkg/(heightinmeter*heightinmeter);
        System.out.println("the BMI (body mass index) is \n " + bmi);
        scanner.close();
    }
}
```

Output:-



Practical:-5

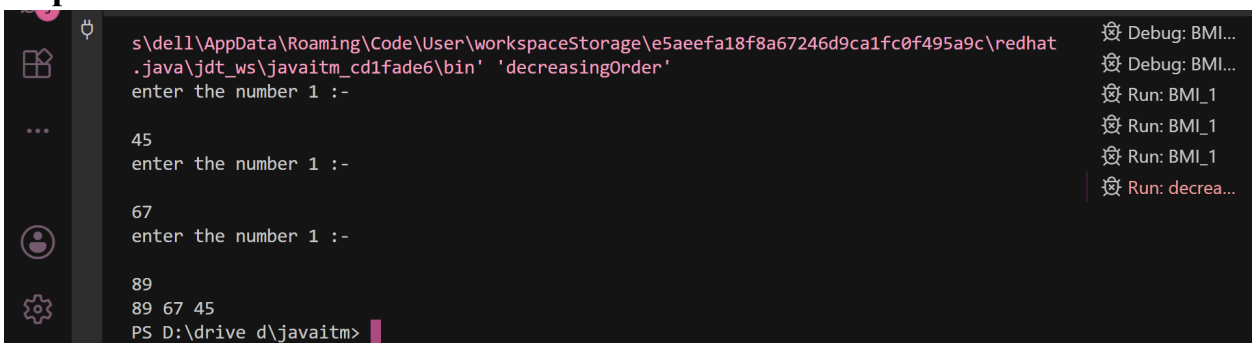
Problem Statement:-

Code:-

```
import java.util.Scanner;
public class decreasingOrder {
    public static void main(String[] args) {
        int a;
        int b;
        int c;
        Scanner scanner = new Scanner(System.in);
        System.out.println("enter the number 1 :- \n  ");
        a = scanner.nextInt();
        System.out.println("enter the number 1 :- \n  ");
        b = scanner.nextInt();
        System.out.println("enter the number 1 :- \n  ");
        c = scanner.nextInt();
        if (a>= b && a>=c){
            if (b>=c )
                System.out.println(a + " " + b + " " +c );
            else
                System.out.println(a+ " " + c + " " + b);
        }
        else if (b>=c && b>=c){
            if(a>=c)
                System.out.println(b + " " + a + " " + c);
            else
                System.out.println(b+" " + c + " "+ a);
        }
        else{
            if(a>=b)
                System.out.println(c+" "+a+" "+ b);
            else
                System.out.println(c+" "+ b+ " "+a);
        }
        scanner.close();

    }
}
```

output:-



Practical:-6

Problem Statement:-Write a program that prompts the user to enter a letter and check whether the letter is vowel or consonant

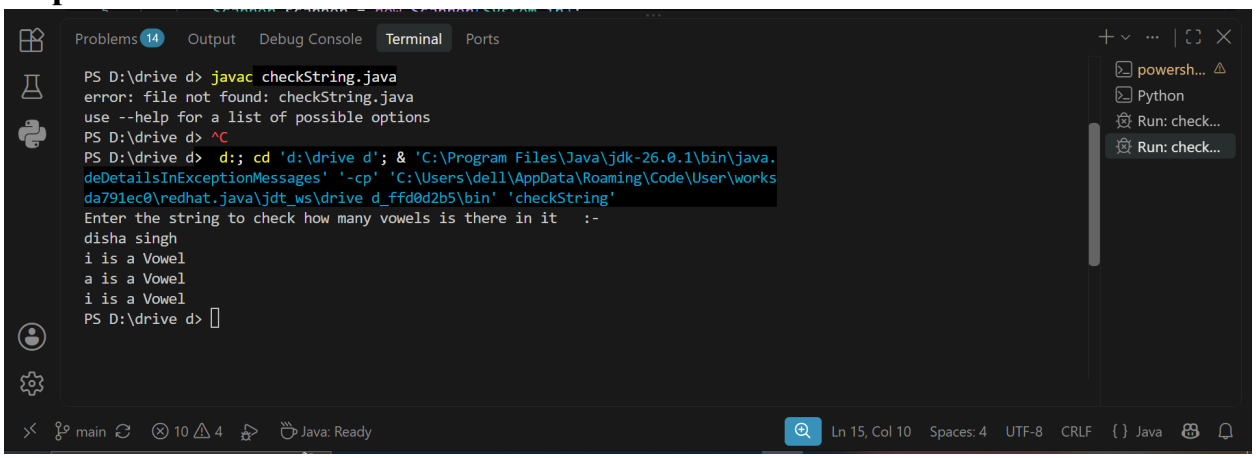
Code:-

```
import java.util.Scanner;

public class checkString {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("Enter the string to check how many vowels is there in it   :- ");
        String str  = scanner.nextLine();

        for(int i = 0; i < str.length(); i++){
            char ch = str.charAt(i);
        if (ch == 'a' || ch == 'e' || ch =='i' || ch == 'o' || ch == 'u' || ch == 'A' || ch == 'E'|| ch == 'I'
            || ch == 'O' || ch == 'U'){
                System.out.println(ch + " is a Vowel");
            }
        }
    }
}
```

output:-



Practical:-7

Problem Statement:- Generate Vehicle plate number consist of three upper case followed by 4 numbers WAP to generate a number plate .

Code:-

```
import java.util.Random;

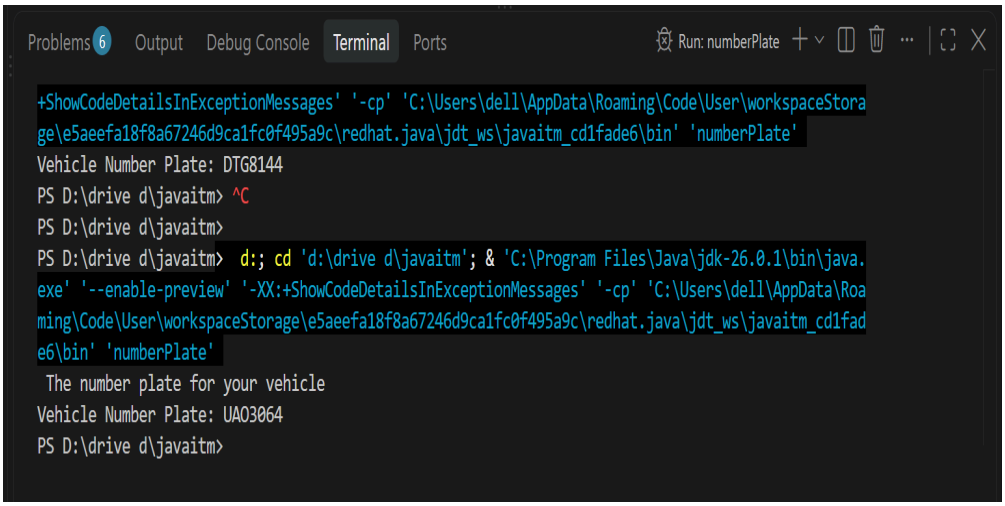
public class numberPlate {
   public static void main(String[] args) {

      Random r = new Random();

      char a = (char)('A' + r.nextInt(26));
      char b = (char)('A' + r.nextInt(26));
      char c = (char)('A' + r.nextInt(26));

      int num = 1000 + r.nextInt(9000);
      System.out.println(" The number plate for your vehicle ");
      System.out.println("Vehicle Number Plate: " + a + b + c + num);
   }
}
```

OUTPUT:-



Practical:-8

Problem Statement:-8. Write a program that reads an integer and displays all its smallest factors in increasing order. For
example, if the input number is 120, the output should be as follows: 2,2,2,3,5.

Code:-

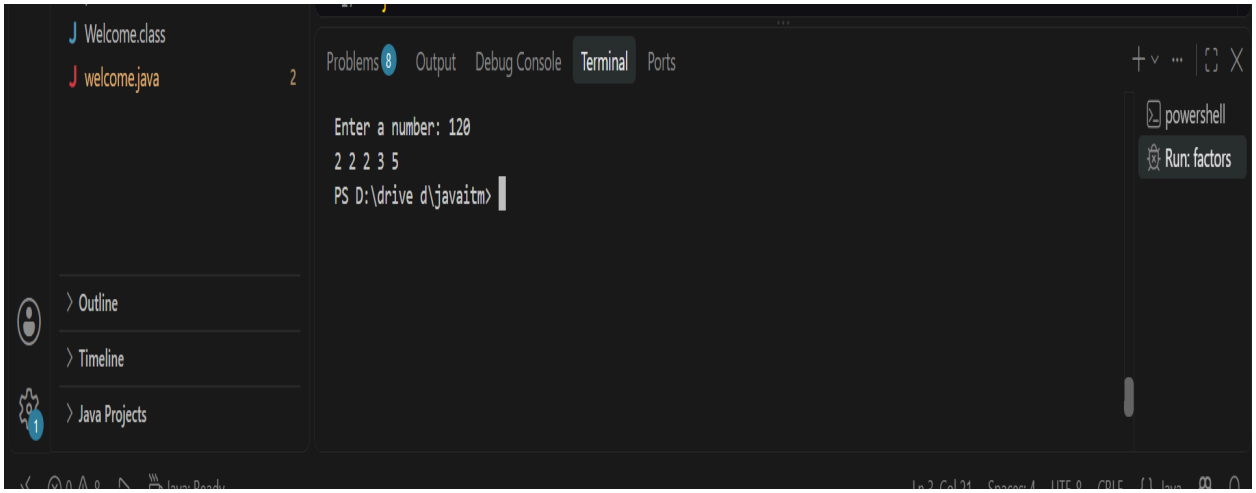
```
import java.util.Scanner;

public class factors {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 2; i <= n; i++) {
            while (n % i == 0) {
                System.out.print(i + " ");
                n = n / i;
            }
        }
    }
}
```

OUTPUT:-



Practical:-9

Problem Statement:- Write a method with the following method header. public static intgcd(int num1, int num2)
Write a program that prompts the user to enter two integers and computes the gcd of the two integers.

Code:-

```
import java.util.Scanner;

public class gcd_Num {

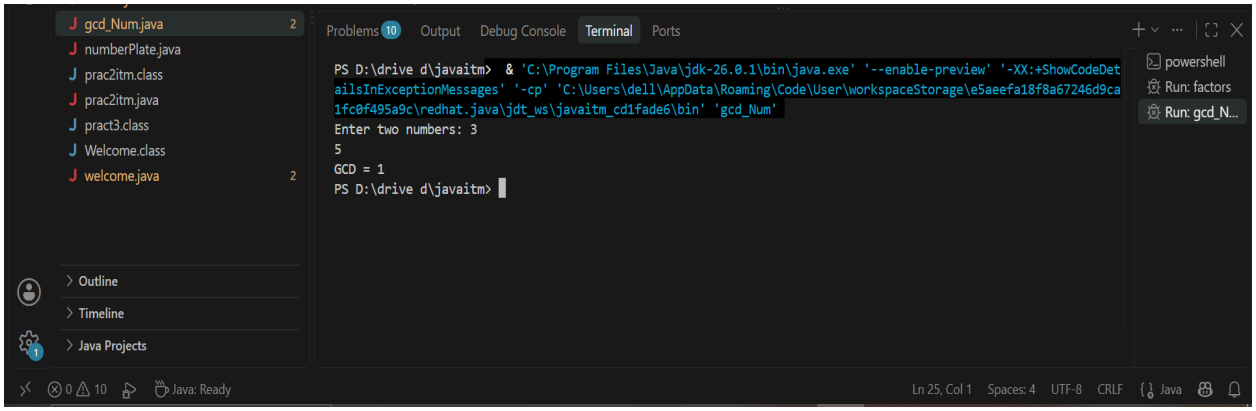
    public static int gcd(int num1, int num2) {
        while (num2 != 0) {
            int temp = num2;
            num2 = num1 % num2;
            num1 = temp;
        }
        return num1;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter two numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println("GCD = " + gcd(a, b));
    }
}
```

OUTPUT:-



Practical:-10

Problem Statement:- Write a test program that prompts the user to enter ten numbers, invokes a method to reverse the
numbers, display the numbers.

Code:-

```
import java.util.Scanner;

public class reverse_Num {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[10];

        System.out.println("Enter 10 numbers:");

        for (int i = 0; i < 10; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("Numbers in reverse order:");

        for (int i = 9; i >= 0; i--) {
            System.out.print(numbers[i] + " ");
        }
    }
}
```

OUTPUT:-

